

Network Forensics Roadmap

Week 1: Computer Basics & System Exploration

Goal: Understand how computers and operating systems work.

Topics:

- Basic computer hardware (CPU, RAM, storage)
- File systems and directories
- Basic command line usage
- Viewing running processes

Activity:

- Explore system commands in
 - Microsoft Windows (Command Prompt)
 - Linux (Terminal)

Outcome:

Understand how systems operate and where digital evidence may exist.

Week 2: Networking Fundamentals

Goal: Learn how devices communicate on a network.

Topics:

- What a network is
- IP addresses
- Ports and protocols
- DNS and HTTP/HTTPS
- Basic OSI model overview

Activity:

- Use network commands to inspect your network
- Identify your:
 - IP address

Network Forensics Roadmap

- Gateway
- Connected devices

Optional practice platform:

- Beginner challenges on picoCTF related to networking basics.

Outcome:

Understand how data travels across networks.

Week 3: Introduction to Cybersecurity & Forensics

Goal: Learn how cyber incidents happen and how investigators analyze evidence.

Topics:

- Types of cyber attacks
- Digital forensics fundamentals
- Evidence collection
- Chain of custody

Activity:

- Complete beginner challenges on picoCTF (Forensics category)

Examples of tasks:

- Decode hidden messages
- Analyze simple files
- Extract information from images

Outcome:

Learn basic investigative thinking used in cybersecurity.

Week 4: Network Traffic Analysis

Goal: Start observing real network traffic.

Network Forensics Roadmap

Topics:

- Packet capture
- Network monitoring
- Identifying protocols in traffic

Tools to try:

- Wireshark
- tcpdump

Activity:

- Capture your own network traffic
- Identify:
 - DNS requests
 - Website connections
 - Packet details

Optional challenge:

- Solve network-related problems in picoCTF.

Outcome:

Learn how investigators capture and analyze network data.

Learning Flow

Week 1 – Computer Basics + Command Line Practice

Week 2 – Networking Fundamentals + Basic Network Commands

Week 3 – Cybersecurity & Forensics Introduction + picoCTF Challenges

Week 4 – Network Traffic Analysis + Wireshark Packet Capture